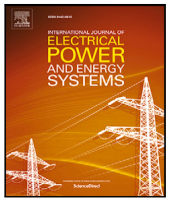
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Denoising autoencoder based topology identification in distribution systems with missing measurements

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ABSTRACT

In distribution systems, the loss of existing real measurements and the stochastic penetration levels of renewable energy sources (RES) are the major issues while identifying the correct topology. The frequent switching and control actions of RES can lead to many topology changes. The loss of existing measurements from the physical meters due to failures in meters or communication aggravates the topology identification process and also it brings the convergence issues in state estimation algorithm. In this work, a deep learning approach is utilized to address the issue of incomplete measurement data and to provide accurate topology identification. A temporal convolutional denoising autoencoder (TCDAE) is developed for dimensionality reduction, reconstruction of incomplete data, and accurate topology identification. The model performance is compared with other traditional deep learning approaches such as simple denoising autoencoder (DAE), convolutional DAE, and the LSTM-based DAE. The power injection measurements and the line current sensor measurements are considered as the input measurement data for topology identification. The simulations have been carried out on IEEE 13-node and IEEE 37-node distribution test systems. The corresponding results indicate that the model can fill the incomplete data with more accurate values and its classification results prove that the model performance is effective in identifying the correct topology.

1. Introduction

Distribution systems are traditionally responsible for feeding the transmitted power to various kinds of consumers. However, the arrival of renewable energy sources (RES) has changed the passive consumers into prosumers. The continuous tracking of dynamic changes in penetration levels of RES is a challenging task for the distribution management system (DMS). Hence, the distribution system state estimation (DSSE) plays an important role in the monitoring of complex distribution systems and the dynamic behavior of RES. Apart from this, the topology analysis is also one of the essential tasks of DMS because the accurate topological input data is the basis of the DSSE algorithm, power flows, network reconfiguration, demand response, etc. [1]. Therefore, the solution of the DSSE algorithm entirely depends on the accurate topology identification (TI) along with the measurement data. The state estimation (SE) gets the configuration data from the network topology processor (NTP) which identifies the correct topology by leveraging the status of switching devices and distributed energy resources (DERs) [2,3]. However, the availability of limited measurements and the unreliable information of the status of switching devices and DERs are the major issues for accurate TI. To achieve accurate TI, different techniques have arrived among which the development in machine learning (ML) algorithms and deep learning (DL) models is the current state of the art.

As the manual tracking of continuous changes in topology is a difficult task, most of the research works have been developed by leveraging the measurement-driven methods that examine the status of switches based on existing meters and sensors [4]. A common approach is to perform SE by utilizing the existing measurements and then, the execution of residual analysis reported in [5] provides the way to identify the branch outage. However, the assumption of SE convergence is not definite because the impact of the topology errors affects the measurement residuals. To address this issue, another approach is developed in which the TI problem is integrated into the SE problem formulation that augments the state vector to provide information of both the system states and the status of the switches [6]. Though this method improves the computational efficiency, the loss of measurement information affects the convergence of the state estimation. The TI and bad data processing (BDP) are performed by using a fuzzy-based c-means clustering technique [7]. However, the increase in the size of the system and the number of possible topologies expands the fuzzy pattern vector which affects the accuracy of TI and state estimation. In [8],

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the node connections are analyzed by leveraging the correlation coefficients of the voltage sequence between nodes. Using the measurement residuals of the weighted least squares (WLS) method, a mixed-integer quadratic programming technique has been formulated in [9] for TI. Another approach is developed in [10] to detect any topology change where an event-triggered TI stage is added to the SE process by using a recursive Bayesian approach. However, the computational time of these methods is quite high for distribution system applications.

The topology identification using voltage measurements has been widely employed in the literature. In [11], the voltage covariance matrix has been used for determining the connection between nodes. The authors in [12] have introduced a time-series signature verification method to detect the connections of nodes by utilizing synchronized voltage phasors. In addition to this, the correlations between different node voltages are used in [13] to identify the topology. In another approach of [14] with the integration of DERs, the system topology can be identified by leveraging the smart inverters which collect and process the voltage deviations across the nodes. Apart from the voltage measurements based TI, the consumer loads and line power flows have been used in [15] to identify the node connections. However, this method needs CTs and PTs to be installed at each measurement point to measure the power flows and hence, the method is expensive.

With the advancements in ML algorithms and DL models, TI can be processed by training them using a large number of datasets. In [16,17], a deep neural network (DNN) is trained for state estimation using power injection measurements. A multi-label classification approach is developed in [18] by identifying the switch actions. In this, the dynamics of DERs and three-phase voltage magnitudes along with frequency are used for training a convolutional neural network (CNN). Also, a DNN approach is applied in [2] for online TI by using the power injection and power flow measurements where dynamic topology detection is studied under the fault condition. In [1], a principal component analysis (PCA) is utilized for dimensionality reduction whose output is fed to the deep belief network (DBN) surrogate model, where the status of tie-line switches and voltage measurements are considered for the TI with the inclusion of DERs. However, the assumption of using node voltage magnitudes as measurements is an expensive method because they are usually obtained from PMUs in real-time. Hence, the line current measurements can be utilized for the topology processing as they can be obtained from the non-contact line current sensors. These sensors have been developed recently for the monitoring of power distribution lines because they are low-cost devices and can be installed easily when compared to voltage and power measurements obtained from traditional devices. The currents above 1 mA can be measured with these line current sensors by considering 1% magnitude error and 1.5° phase angle error for every second-second reporting rate [19]. These sensors measure currents based on electric field and hence, they are unable to measure voltage because of which power measurements are also not possible to measure [4].

As the presence of missing samples in the data affect any data-based applications, the recent literature includes the use of ML algorithms and DL models for imputing the missed data. In [20], a learning adaptive imputation method (LAI) is developed for missing data, where the k-Nearest Neighbors (kNN) algorithm is applied on the optimal length of historical data. In [21], the model is developed by leveraging the K-means clustering algorithm where the different sizes of historical data and clustering distance are used for analysis. A stacked denoising autoencoder (DAE) model is developed in [22] where the loss of monitoring data in smart grids are imputed. Another DL-based approach namely, generative adversarial network (GAN) is utilized in [23] which can be used for filling the incomplete data by identifying the probability distribution in the dataset. In [24], the authors have analyzed the imputation of missing samples for smart meters by leveraging a DAE framework. This model can understand the correlation among the input data but fails to capture the temporal information.

In the above literature, the effect of the presence of outliers in the input data is not discussed while filling the incomplete data. The presence of outliers is the major problem which significantly affects the performance of most of the ML and DL models during the imputation of missing data. Although the DAE framework is effective in replacing the outliers and missing samples with the reconstructed data, it cannot capture both spatial and temporal dynamics simultaneously from the input data. The accurate TI is an inevitable task in DSSE and the loss of existing measurements from different branches in the system may lead to inaccurate TI. Hence, there is a requirement of suitable model which can replace the missing samples and outliers with more accurate values to achieve the effective results for TI and DSSE.

In this work, a novel DL approach called temporal convolutional denoising autoencoder (TCDAE) is proposed which can extract both temporal and spatial features from input data to reconstruct the presence of missing data and outliers. This model can effectively reconstruct the incomplete measurements and outliers by learning the correlations among the spatio-temporal features of input data. Further, the addition of an autoencoder structure gains the advantage of dimensionality reduction. This model leverages the dilated causal convolutions for the tracking of long-term information by increasing the receptive field. In self-supervised mode, the proxy task is performed by feeding the noisy measurement data (missing samples and outliers) to the TCDAE model for the reconstruction of actual measurement data. For this task, a modified Huber-loss function is proposed by incorporating the standard deviation of measurements to get the imputed values within the maximum error that can be minimized with the DSSE algorithm. Later, the robustness of the model is evaluated in classification mode to detect the accurate topology with different percentages of missing samples and outliers. For this, the multi-label classification is employed in supervised mode by considering the focal loss function which can eliminate biasing problems towards easily classified examples. The magnitudes of line current flows and load power injections are assumed to be available as measurement data in this work. Finally, the performance of the proposed model is evaluated with the classification metrics and compared with a simple denoising autoencoder (DAE), Convolutional DAE (CDAE), and LSTM-based DAE (LDAE) under the presence of different percentages of missing data, and outliers. From the comparison results, the proposed TCDAE model achieved better performance in TI. Also, the DSSE is performed with and without TI for both test systems, and it is evaluated with a two-norm error metric.

The rest of the paper is structured as follows: The background of the TI problem and measurements are discussed in Section 2. The workflow of the proposed TI along with DSSE, the frameworks of DAE, CDAE, LDAE, and the structure of the proposed TCDAE are described in Section 3. The simulation data description of the test systems, DAE based DL models, the simulation results of reconstruction of missing samples and outliers, topology classification, and the DSSE performance with and without TI are reported in Section 4. The summary of work is finally concluded in Section 5.

2. Overview of the topology identification (TI)

In general, the transmission network is mainly operated in mesh configuration and its topology change occurs very rarely. However, the topology changes can be observed more frequently in the case of medium voltage (MV) and low voltage (LV) distribution systems and hence, it is a challenging task to detect a correct topology in distribution system using classical methods [1].

2.1. TI problem formulation

In practice, the distribution networks are usually designed in a closed-loop manner but they operate in the open-loop form to maintain the radial connection by ensuring reliability, security, and simplicity. The reliability and security can be guaranteed by operating the tie-line

switches, capacitor banks, and sectionalizing switches. So, the distribution system topology can be identified by observing the connectivity between the nodes with the help of binary 0/1 status of tie-switches and sectionalizing switches. For instance, all the lines and nodes are in service to form a radial connection with the help of tie-switches. Now, by applying Kirchhoff Current Law (KCL), the current injection to each node is given by

$$I_i = \sum_{j \in \mathcal{N}_i} I_{ij}, \quad i \in \mathcal{N} \tag{1}$$

where $\mathcal{N}$ is the number of nodes and I_{ij} is the branch current connecting the nodes i and j. This branch current mainly depends on the status of line switches, the voltage of the two nodes, and the branch admittance. Now, the Kirchhoff Voltage Law (KVL) can be applied:

$$I_{ij} = s_{ij}(V_i - V_j)y_{ij} \tag{2}$$

where s_{ij} is the switch status of branch i-j and y_{ij} is an admittance of branch i-j, V_i and V_j are the voltages at nodes i and j respectively. If branch i-j is in service, then s_{ij} is considered as 1; otherwise, it is treated as 0 which represents that the branch is out-of-service. With this equation, the TI can be processed by utilizing either the measurements of branch currents/power flows or the voltage measurements. With the advent of cheap non-contact line current sensors, the magnitudes of branch current flows are utilized in this work to identify the topology. The main advantage of using the line current measurements is that the model can classify the correct topology even in the presence of bad data. Hence, the topology can be easily identified based on the line connectivity status which is given by the presence of line current measurement. In addition to the line current measurements, the power injection measurements are also considered. However, the loss of measurements at any instance leads to misidentification of topology which affects the convergence or the solution of DSSE. To address these issues, a data-driven DL approach is exploited for dimensionality reduction, reconstruction of missing data, and accurate TI.

2.2. Measurements

In this work, the noisy measurements are represented as follows:

$$\hat{z} = z + \eta \tag{3}$$

where $\hat{z}$ denotes the erroneous measurement vector, z is the true measurement vector and η is the noisy vector added to the true measurements. The noise is assumed to follow the normal distribution with zero mean value and $\pm 3\sigma$ deviation about the mean by which the error value falls within 99.73% of the area. For a given percentage of maximum error about the mean μ_{z_i}, the standard deviation σ_{z_i} of the error can be determined by [25]

$$\sigma_{z_i} = \frac{\mu_{z_i} \times \%error}{3 \times 100} \tag{4}$$

In this work, the following measurements are considered:

1. Line current sensors: These sensors are non-contact devices used to measure the branch current flows in the system. The error in line current magnitudes is considered as 1%.
2. Smart meters (SMs): In practice, the load power measurements are obtained from the smart meters located at consumer premises.
3. Pseudo-measurements: In general, the utility's knowledge about these measurements is uncertain because they are obtained using forecasting methods based on historical data or smart meter data. These measurements are usually active power and reactive power of various kinds of consumers. When the physical existence of metering is not available, these measurements are very helpful for TI and DSSE. Their uncertainty completely depends on the way how the measurements are predicted based on historical data.

3. Description of the methodologies

3.1. Workflow of the DAE-based methods

The description of the DAE based methodology for TI and DSSE is represented with a flowchart shown in Fig. 1. In general, the accurate solution of the DSSE requires sufficient measurement data and parameters data of correct topology. However, the robustness of DSSE has to be tested against all the possibilities of uncertainties that occur in real-time. In this paper, the DAE based DL approach is exploited to reconstruct the incomplete data and outliers present in the measurement data. As the measurement process is usually monitored by the devices located in the outside environment, there is a high risk of losing some of the measurements either due to the failure in communication or the repairs in measurement devices. Sometimes, outliers may exist in the measurements due to the malfunction of the meters or any other reasons. To address these problems, the DAE-based DL methods are trained with the simulated data obtained from power flows in the offline stage. In this offline stage, the noise for the DAE models is created by randomly replacing some of the samples with 0's and outliers. Later, the real-time measurement data is fed to the trained DAE model which reconstructs the missing samples and outliers that exist in the input data. In addition to this, the same model can be used for topology classification by replacing the decoder part with the dense layers. Finally, the entire measurement data can be applied to the DSSE algorithm along with the identified topological parameter data. The modified Huber-loss function is used as a reconstruction loss function to get the reconstructed data within $\pm 3\sigma$ deviation which can be easily handled by the DSSE algorithm.

3.2. DAE methodologies

In this part, different DAE-based deep learning models are discussed for the reconstruction of missing data and outliers. During the reconstruction process, the DAE models are developed using the encoder and decoder structures and they trained in self-supervised learning mode as they do not need any external labeled data. In the classification task, the decoder part of the network is replaced by fully connected layers and the model gets trained in supervised mode.

An autoencoder (AE) is one form of a DL based approach where high dimension measurement data can be encoded into a latent representation with reduced dimension. The applied data can be reconstructed from this latent space using the decoder part of the model. Whereas in DAE, the noise is added to the input before feeding it to the AE to reconstruct the original input. The DAE framework with encoder and decoder layers and their dimensions are illustrated in Fig. 2. In this work, the noise is created by corrupting the original input data x which can be done by randomly placing 0's and outliers, and this corrupted data is denoted as $\hat{x}$. The encoder part of the model encodes the input $\hat{x}$ to latent space z with a nonlinear function $f(\hat{x})$. Later, the decoder part of the model decodes the transformed z to the reconstructed data $\tilde{x}$ by a nonlinear function $g(z)$. Finally, a reconstruction loss is formulated between the reconstructed data and the original data. The model is trained by minimizing the error between x and $\tilde{x} = g(f(\hat{x}))$ to get the reconstructed output $\tilde{x}$ closer to the original input data x [24]. In this way, the generalization capability of DAE is achieved to the partial destruction of input by learning the correlation between the input features and mapping them to an equivalent latent representation. The encoder and decoder layers of a simple DAE model consist of dense layers as represented in Fig. 2. In the case of convolution based DAE, the dense layers of the encoder and decoder parts of the model are replaced by 1-D convolutional layers and transposed 1-D convolutional layers respectively. The kernel size of 1×1 is used for all the convolutional layers. The advantage of adding the convolutional structure in encoder and decoder parts is that it facilitates the enhancement in the performance of the model by capturing the important spatial

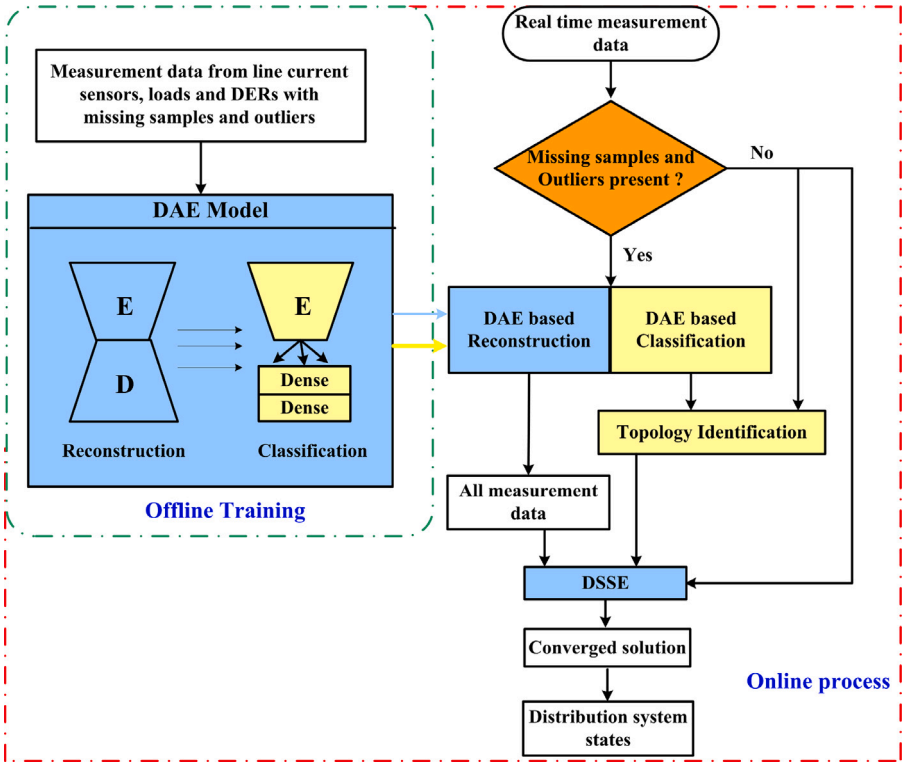


Fig. 1. Workflow of the proposed methodology.

information from the corrupted data. In reconstruction task, the model tries to impute the missing samples and outliers from the corrupted data. Later the model gets trained in supervised mode by flattening the encoder output before applying it to the fully-connected network. An LSTM based DAE is constructed by leveraging the LSTM layers in encoder and decoder parts of the model. The use of LSTM layers in the encoder structure facilitates the model to capture the temporal information from the corrupted input vector and it represents those temporal features into the compressed latent space.

For all the above DAE structures, the same number of filters are used i.e. 32, 20 for the encoder layers and in a transposed manner for the decoder layers. From Fig. 2, the cross marks 'x' and circles 'o' in the boxes represent the missing samples and outliers respectively.

3.3. Proposed methodology

Fig. 3 illustrates the proposed method using TCN based denoising autoencoder with the destructive data created by adding missing samples and outliers in the original input data. In the case of sequential data applications, the convolutional network is generally more effective than the recurrent structures. In this paper, a temporal convolutional network [26] is utilized for the application of real-time measurement data. The TCN model can have different sizes of input and output sequences in the time series forecasting approach. The history of research studies shows that feature learning can be done with the CNN model while the sequential modeling tasks can be dealt with RNN structures. Hence, this work utilizes the TCN model which follows both the RNN and CNN concepts. The TCN model utilizes causal convolution in which the output is independent of the future inputs. The output produced by the TCN model at time t depends only on the past samples i.e. $t-1$, $t-2$, etc. Similar to the RNN structure, the TCN model can produce an output sequence of the same length as that of the input sequence. This can be done with the TCN model for any input sequence length by using a 1-D fully connected layer.

3.3.1. Structure of TCN

The main structure of the TCN consists of the following elements:

Causal convolution: As the name suggests, it allows the model output x_t at timestamp t to be affected by only a few previous inputs at time $t-1$, $t-2$, etc. To use a bigger size of historical data, a large receptive field is required which can be achieved by a larger filter size. However, the use of a larger filter size is the main cause of the non-convergence issues and also, the use of an extensive deeper network brings the problem of model training issues. To alleviate these issues, a dilated convolution with residual block concept is introduced to obtain a larger historical data size.

Dilated convolution: The basic CNN model normally consists of a series of convolution operations followed by pooling operations to compress the size of sequential data with a simultaneous increase in the size of the receptive field. The use of pooling operations causes the model to lose some of the information which is the main disadvantage of the normal CNN model. This problem can be overcome by adopting the dilated convolution in the TCN model, which does not need any pooling operation and it allows the model to encompass the important information from long-term tracking by increasing the perceptron field with a series of dilated convolutions. Hence, the concept of dilated convolutions has huge importance in time series applications where long-term dependencies are important. As the problem involved in this work utilizes the real-time series data for the loads and other measurements, the TCN model is an efficient approach to handle the uncertainties like missing data and outliers. The dilated convolution operation f on the element with a sequence x is represented as [27],

$$f(s) = (x_d^* f)(s) = \sum_{i=0}^{k-1} f(i) x_{s-d,i} \tag{5}$$

where d represents the dilation factor, k is the filter size and s-d,i considers the direction of the past.

Residual block: The other important part of the TCN model is the residual connection [28]. Fig. 4 gives the structure of the residual block which normally consists of dilated causal convolution, weight

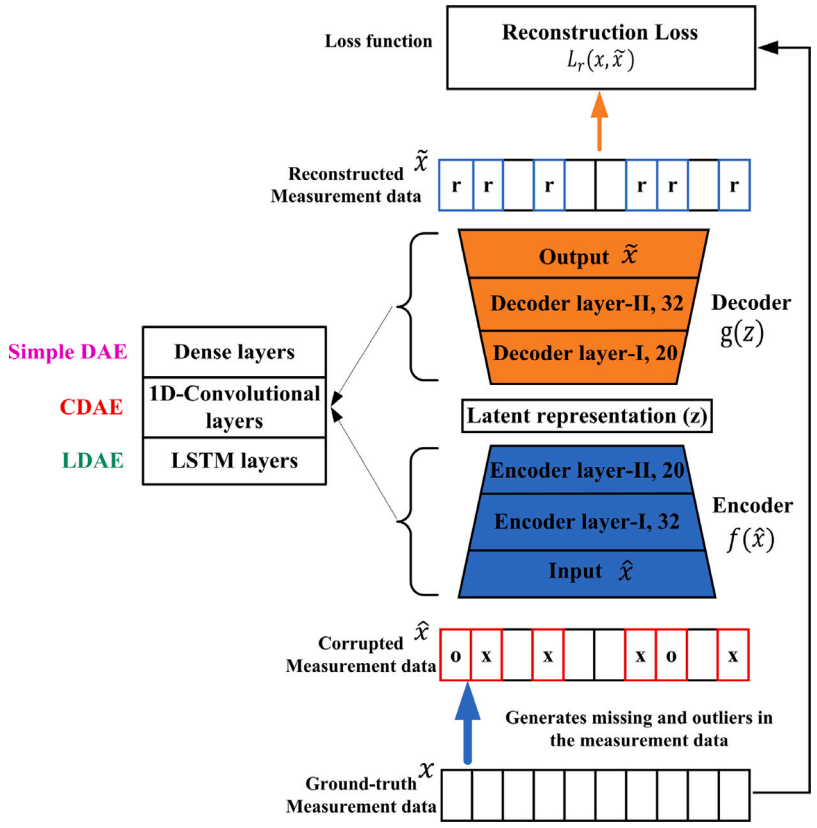


Fig. 2. DAE framework.

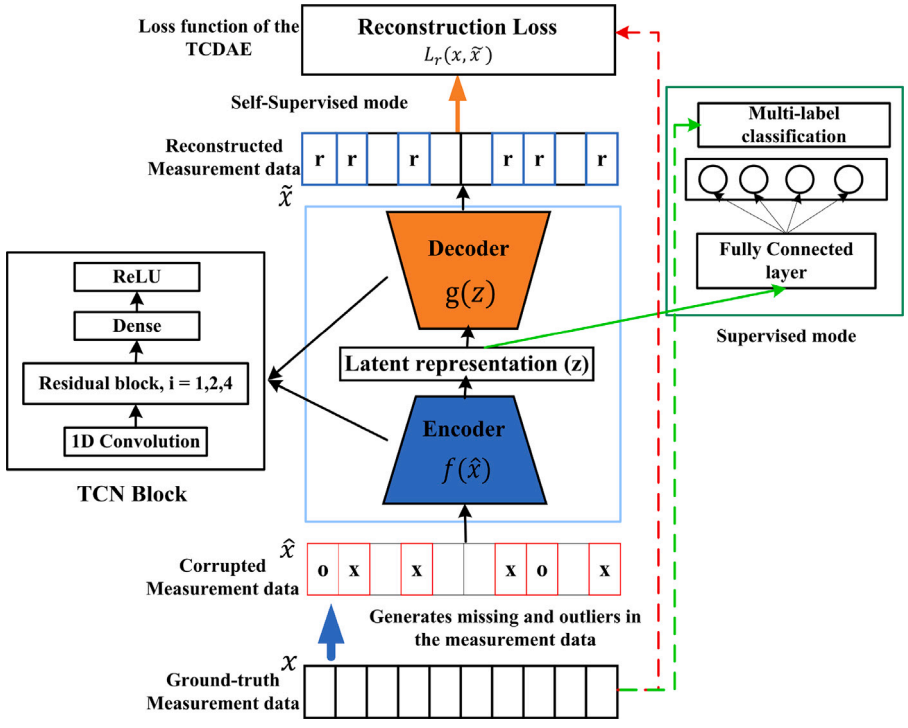


Fig. 3. Framework of TCN-based DAE.

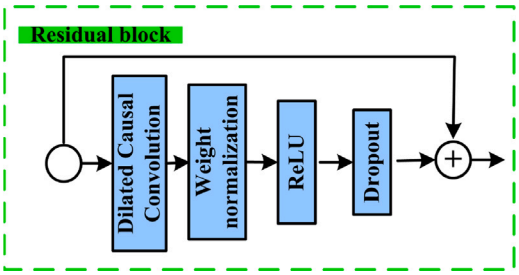


Fig. 4. Residual block of TCN.

normalization with an activation function ReLU and dropout layer. The residual blocks are also called temporal blocks in which several convolutional layers of the TCN blocks are stacked. It eliminates the training issues of the vanishing gradient problem during backpropagation by combining the previous input and output of the respective convolution with the last input. It includes a skip connection which adds the input directly with the output obtained from a series of transformations. So, the TCN structure has a direct connection to conduct residual mapping from x to $f(x)$. This mapping operation can be expressed as,

$$y = \phi(x + f(x)) \tag{6}$$

where ϕ is a nonlinear activation function, x is the input, and y is the output of residual block.

3.3.2. TCN based denoising autoencoder

The framework of the TCN based DAE structure is formulated in Fig. 3. The steps followed by the model are represented below:

1. In this work, the encoder part of the DAE is modeled with a TCN block which encodes the input length of measurement data to the reduced dimension of size 20, decided by the number of filters. A dilation sequence of $i = 1, 2, 4$ is employed in this work. The advantage of adding TCN block in the encoder is that it can capture both the spatial as well as temporal information from the input sequence of the measurement data. Later, the decoder part also consists of a TCN block which reconstructs the compressed data to the original length of input data.
2. The model gets trained in a self-supervised mode by minimizing a reconstruction loss function between original input x and the model output $\tilde{x}$. Since the mean-squared error loss function is sensitive to the presence of outliers, the Huber-loss function is used as a reconstruction loss function which is a combination of mean-squared error and mean absolute error. In this work, the Huber-loss function is modified according to the DSSE problem:

$$L_\delta(\tilde{x}, x) = \begin{cases} (\tilde{x} - x)^2, & \text{for } |\tilde{x} - x| \le \delta \\ \delta|\tilde{x} - x| - \frac{1}{2}\delta^2, & \text{otherwise} \end{cases} \tag{7}$$

   where, δ is decided by the product of standard deviation of the measurement σ and the true value of measurement x. Hence, the proposed method can replace the missing data and outliers with the imputed values which lie within the maximum error that can be handled by the DSSE algorithm. In (7), the first part belongs to the mean squared error (MSE) which is applied when $|\tilde{x} - x| \le \sigma x$ and the second part is related with mean absolute error (MAE) which is applied when the above condition is false. In this way, the proposed method is robust to the presence of missing data and bad data in the input measurements.
3. Once the model is trained in self-supervised mode, the encoded output from the latent representation is applied to the fully-connected network and the model is allowed to train in supervised mode. In this mode, the binary cross-entropy loss is replaced with a focal loss for allowing the model to assign high weightage to the hard, misclassified examples. The focal loss (FL) function for the binary multilabel classification problem is given below:

$$FL_{loss} = -[\alpha(1-p)^\gamma log(p) + (1-\alpha)(p)^\gamma log(1-p)] \tag{8}$$

   where p is the model's predicted probability, α and γ are the hyperparameters selected as 0.25 and 2 respectively. The α is a weighing parameter and γ is a focusing parameter which smoothly adjusts the rate at which easy examples are down-weighted.
4. In this way, the proposed model can be very useful for both the DSSE algorithm and TI. By training the model in self-supervised mode, the reconstructed data can be utilized for the DSSE algorithm to maintain observability under the presence of missing samples and also, the bad data can be replaced with the reconstructed data. Further, the same model can be utilized for the topology classification after training it in supervised mode.

The procedure of work done by the proposed TCDAE model is represented in Algorithm 1.

Algorithm 1 Steps followed by the TCDAE model

Proxy task for self-supervised mode

Input: The actual input measurement data x and the noisy level is α in %;

Output: The reconstructed measurement data $\tilde{x}$;

Steps

1: Initially, the dataset x is divided into train set x_{train} and testing set x_{test};
2: The noisy input data $\hat{x}_{train}$ is generated by corrupting the actual data x_{train} with the addition of 0's and outliers at random locations given by uniform distribution; $\hat{x}_{train} = N.^* x_{train}$, where N is the noise matrix with 1's and 0's. The presence of 0's represents the missing samples and outlier locations with a percentage of α;
3: Apply $\hat{x}_{train}$ as the noisy input data to the TCDAE model and feed true measurement data x_{train} at the output of the TCDAE model;
4: The $\hat{x}_{train}$ is encoded into latent representation by understanding the correlations between the inputs which is labeled as z; The encoded output z gets reconstructed at the output which is labeled as $\tilde{x}$;
5: With the help of the reconstruction loss function i.e. modified Huber-loss function (Eq. (7)), the missing samples and outliers get reconstructed closer to actual measurements. In this way, the parameters of the encoder and decoder structures get trained to minimize the loss function;
6: Later, the noisy test data $\hat{x}_{test} = N.^* x_{test}$ is fed to the model for the validation of the trained model.

Supervised mode for classification task

Input: The actual input measurement data x and the noisy level is α in %;

Output: Multi-label classification outputs $(\tilde{y})$;

Steps

1: The actual output labels (y) also get divided into y_{train} and y_{test} corresponding to input data x_{train} and x_{test};
2: Connect the MLP network to the output of the encoder structure by excluding the decoding structure. Apply $\hat{x}_{train}$ as the noisy input data to the TCDAE model and feed actual output labels y_{train} at the output;
3: With the help of the focal loss function (Eq. (8)), the topology identification task is performed by minimizing the loss between actual outputs y_{train} and predicted outputs $\tilde{y}_{train}$;
4: Finally, the noisy test data $\hat{x}_{test}$ is fed to the model and the model's performance is evaluated with the classification metrics for TI.

4. Simulations and results

The simulation studies are performed on modified IEEE 13-node (test system-1) and IEEE 37-node distribution (test system-2) systems using MATLAB environment. The simulations are carried out on Intel(R) Core(TM) i7-4790 CPU, 3.60 GHz, 4.00 GB RAM personal computer.

4.1. Data description

The proposed model's performance on the imputation of missing data and topology identification is evaluated by using the real-time power consumption data of commercial, residential, and industrial loads along with power injection data of renewable energy sources (RES) have been collected from National Renewable Energy Laboratory (NREL). Then, the dataset is prepared on an hourly basis by considering the 50 days (i.e. 1200 h) of NREL data. These data are normalized in accordance with the scale of active power (P) demands of the IEEE test systems. The reactive power (Q) is calculated by determining the power factor of steady-state P and Q values of the respective IEEE test systems. Finally, the measurement datasets are generated by feeding the node power data P and Q to the distribution load flow algorithm. Later, the power injections and line current magnitudes are considered as measurement datasets for the TI. The DL models considered in this work are developed using the Keras framework. In this work, some of the power injection measurements are assumed to be obtained from smart meters and rest of them are considered as pseudo-measurements, and line current measurements are obtained from sensors. The noise for SMs is chosen as 10% and the noise for the pseudo measurements is selected based on the uncertainty involved in the prediction such that 30% for residential consumers, 40% for commercial loads, and 50% for industrial consumers [29]. While, the noise for the line current magnitudes obtained from the sensors is considered as 1%. After corrupting the measurements with their respective noise levels, the measurement datasets are fed to the DL models and accordingly, the performance of the models is evaluated. The CNN, LSTM and TCN models are constructed with encoding and decoding structure and their reconstruction of original data from the latent representation are validated with different percentages of missing data and outliers.

4.1.1. Distribution system setup

The unbalanced IEEE 13-node and IEEE 37-node distribution systems are considered as test system-1 and test system-2 respectively and these systems are modified as follows: (a) The presence of transformers and regulators are omitted and the distributed loads are considered as lumped loads. (b) In the case of the test system-1, the switchable lines are assumed to be connected between nodes 2–3, 3–8, 3–11, 7–11, and 10–4. While in the test system-2, the switchable lines are assumed to be located between nodes 9–34, 14–5, 17–30, 26–32, and 32–20. The switches are considered for these lines to maintain the characteristics of radial distribution without forming meshes and islanding of any node. With these assumptions, a total of 8 different configurations for test system-1 and a total of 25 configurations for test system-2, are formulated such that the removal of any fixed line due to faults or maintenance purposes should not isolate any node in the system. (c) In order to check the effect of power injections from RES on topology detection, the nodes 4th and 11th are assumed to be located with photovoltaic (PV) systems in test system-1 while in the case of test system-2, the PV systems are assumed to be located at 18th, 24th, 25th, 26th, 30th, 32nd, 35th nodes along with the wind turbines (WTs) located at 4th, 8th, and 29th nodes. The test system-1 and test system-2 after including the above modifications are represented in Figs. 5 and 6. The switching lines are assumed to be closed such that all the nodes should maintain the continuity of supply under the presence of a fault in any fixed line. With this assumption, the topology scenarios are formulated and the status of switching lines in the ON condition for each topology is represented in Table 1.

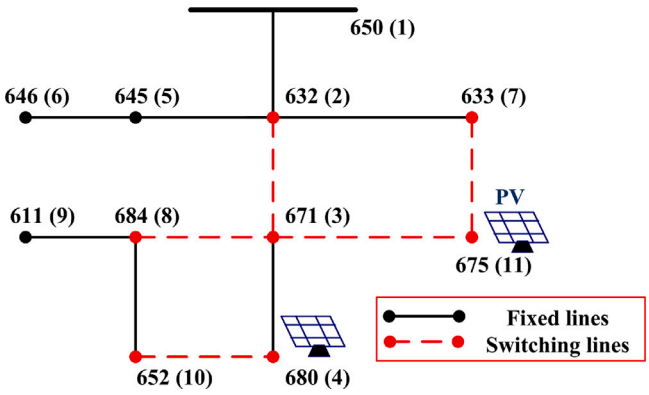


Fig. 5. Single line diagram of modified IEEE 13-node distribution test system.

4.1.2. DAE based DL approaches for topology identification

In the case of the test system-1, there are 17 power injection measurements and 27 measurements of branch current flow magnitudes. The number of topologies formulated for this system is 8, with each topology consisting of 1200 h of data. Hence, the DAE model's input dimension is 44 neurons with 9600 datasets. A 4-digit binary number is assigned to each topology for classification task and hence, 4 outputs are used for the test system-1. In the case of the test system-2, there are 41 power injection measurements and 114 branch current magnitude measurements. In this system, the number of topologies configured is 25, with each topology consisting of 1200 h of data. Thus, the DAE model's input size is 155 neurons with 30 000 datasets and a 5-digit binary number is assigned to each topology representation.

For both the test systems, the same number of hidden layers and sizes are used which are mentioned in Figs. 2 and 3. The ReLU activation function is used for all the layers except for the decoder output layer which uses the linear activation function in the reconstruction task. Whereas in the case of the classification task, the fully connected layer at the output employs the sigmoidal activation function. The modified Huber-loss function is used for all the DAE models during the reconstruction process, the binary focal loss function is utilized during the classification task and their details are discussed in Section 3.3. The Adam optimizer is employed with a learning rate of 0.001 and a batch size of 32 is considered for the training of all DAE models. Before feeding the datasets to the DAE-based models, the dataset is split into 80% for training and 20% for testing. Later, the dataset is normalized using a Min-Max scaler and finally, the data is corrupted by placing the 0's, and outliers which are created by replacing some of the samples with 150% of original values. These missing samples and outliers are placed at random locations which are obtained from a uniform distribution. A dense layer of 100 neurons is connected to the encoded output of the models for collecting the latent features and to produce the equivalent 4-digit (IEEE 13) or 5-digit binary number (IEEE 37).

4.2. Reconstruction of original data with the presence of missing samples and outliers

In this section, the DAE-based DL models are formulated and tested by creating the various levels of corrupted data. The corrupted data consist of both missing samples and outliers which significantly affects the topology detection and DSSE algorithm. The missing samples and outliers are created at random locations to generate the noisy measurement data for the topology detection.

The Root Mean Square Error (RMSE) metric is used for comparing the DAE, CDAE, LDAE, and TCDAE. Also, one of the efficient machine learning algorithms for missing data namely, the k-nearest neighbors

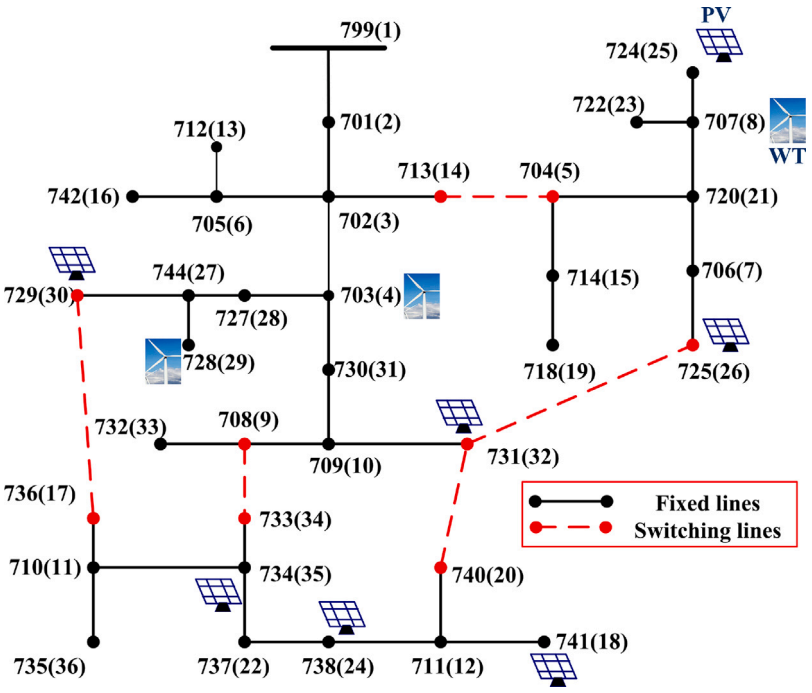


Fig. 6. Single line diagram of modified IEEE 37-node distribution test system.

Table 1
Switching lines in service for each topology.

Test system-1		Test system-2	
Fault in branch (topology number)	Switching lines in service	Fault in branch (topology number)	Switching lines in service
Healthy condition (1)	2-3, 3-11, 3-8	Healthy condition (1)	14-5, 9-34
8-10 (2)	7-11, 11-3, 3-8, 4-10	3-14 (2)	5-14, 32-26, 9-34
8-10 (3)	2-3, 3-8, 4-10, 7-11	5-21 (3)	14-5, 32-26, 9-34
8-10 (4)	2-3, 3-8, 4-10, 3-11	21-7 (4)	14-5, 32-26, 9-34
3-4 (5)	2-3, 3-8, 10-4, 7-11	7-26 (5)	14-5, 32-26, 9-34
3-4 (6)	2-3, 3-8, 10-4, 3-11	3-4 (6)	14-5, 26-32, 9-34
3-4 (7)	7-11, 11-3, 3-8, 10-4	4-28 (7)	14-5, 9-34, 17-30
2-7 (8)	2-3, 3-11, 11-7, 3-8	28-27 (8)	14-5, 9-34, 17-30
–	–	27-30 (9)	14-5, 9-34, 17-30
–	–	4-31 (10)	14-5, 9-34, 26-32
–	–	4-31 (11)	14-5, 30-17, 34-9
–	–	31-10 (12)	14-5, 9-34, 26-32
–	–	31-10 (13)	14-5, 30-17, 34-9
–	–	10-32 (14)	14-5, 9-34, 26-32
–	–	10-32 (15)	14-5, 9-34, 20-32
–	–	10-9 (16)	14-5, 30-17, 34-9
–	–	10-9 (17)	14-5, 32-20, 34-9
–	–	34-35 (18)	14-5, 32-20, 9-34
–	–	34-35 (19)	14-5, 30-17, 9-34
–	–	35-11 (20)	14-5, 30-17, 9-34
–	–	11-17 (21)	14-5, 30-17, 9-34
–	–	35-22 (22)	14-5, 32-20, 9-34
–	–	22-24 (23)	14-5, 32-20, 9-34
–	–	24-12 (24)	14-5, 32-20, 9-34
–	–	12-20 (25)	14-5, 32-20, 9-34

(kNN) approach is implemented (with neighbors = 5) to compare with the DAE-based approaches. The equation for RMSE is represented by

$$\text{RMSE} = \sqrt{\frac{1}{n}\sum_{i=1}^{n}(\tilde{x}_i - x_i)^2} \tag{9}$$

where, n is number of missing samples and outliers; $\tilde{x}$ is reconstructed output; x is the actual data without missing samples and outliers. Table 2 demonstrates the RMSE of DAE approaches and kNN at different percentages of missing data with the presence of 30% outliers. Based on the results, the proposed TCDAE method is effective compared to other approaches as it captures both spatial and temporal information from the data. Since the data used in this work is related to hourly time series data, the LSTM-based DAE is the second best model in the reconstruction of missing samples and outliers. The LSTM cell is having the capability to capture the long-time dependency in the data which allows the model to work effectively in reconstruction. The CNN-based DAE is also effective as it captures the spatial information and also it can extract the important features from the applied data. As TCDAE is having the capability of understanding both temporal and spatial information, it is more effective compared to the other models.

For the test systems, the effectiveness of the proposed TCDAE model is depicted in Figs. 7 and 8 with 10% missing samples and 30% outliers. The power injection and line current sensor measurements at a particular time instance are considered for Figs. 7 and 8. The

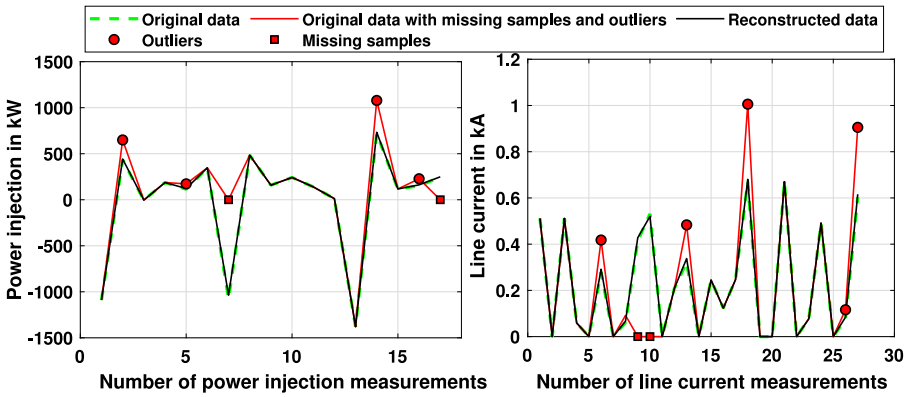


Fig. 7. Power injection and line current measurements of test system-1 with missing samples and outliers.

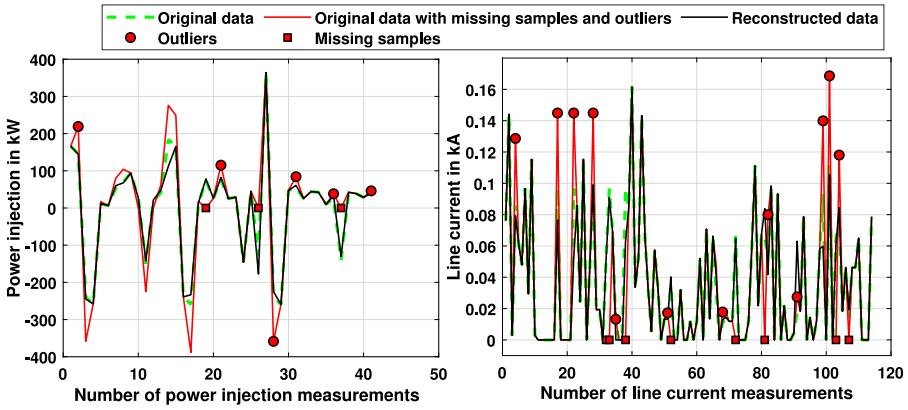


Fig. 8. Power injection and line current measurements of test system-2 with missing samples and outliers.

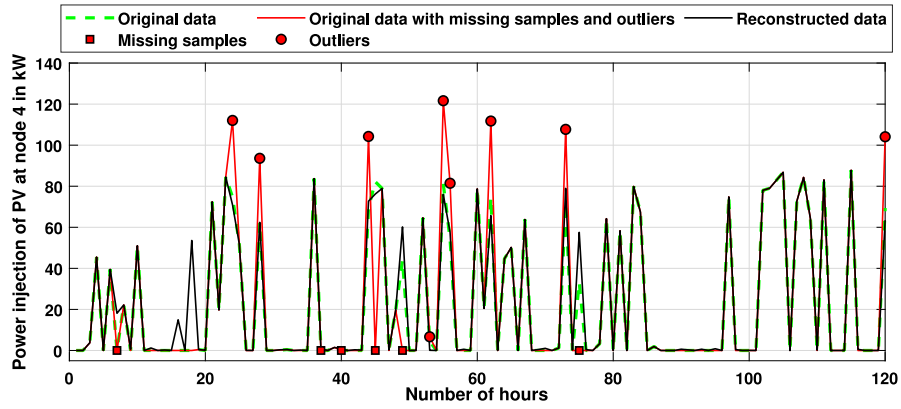


Fig. 9. PV power injection at node 4 (5 days data) of test system-1 with missing samples and outliers.

Table 2
RMSE of DAE-based DL models and kNN with different percentages of missing data.

Model	RMSE with different % of missing and 30% outliers					
	Test system-1			Test system-2		
	10%	30%	50%	10%	30%	50%
DAE	18.53	22.57	26.53	9.75	13.35	17.50
CDAE	17.54	20.93	24.93	9.60	12.93	17.23
LDAE	17.28	20.64	24.83	9.45	12.54	16.98
TCDAE	**16.24**	**19.39**	**22.44**	**8.10**	**10.75**	**15.23**
kNN	27.43	30.41	34.41	25.39	27.47	29.17

reconstruction of missing samples and outliers by the proposed model is almost following the original values which are present at missing and outlier locations. Fig. 9 is indicating that the power injection of the PV system at node 4 in test system-1 by considering the 120 h of testing data. Similarly, Fig. 10 shows the power injection of the wind turbine at node 4 in test system-2 for the 120 h of testing data. Since the PV and WT power generations are intermittent in nature, the exact reconstruction of missing samples and outliers is not possible. However, the reconstruction values of the proposed model are not significantly deviating from the original values. The presence of missing samples and outliers significantly affects the TI and the solution of DSSE. Hence, the DAE can be used to reconstruct the missing samples and outliers closer to actual values and then, these reconstructed data along with other

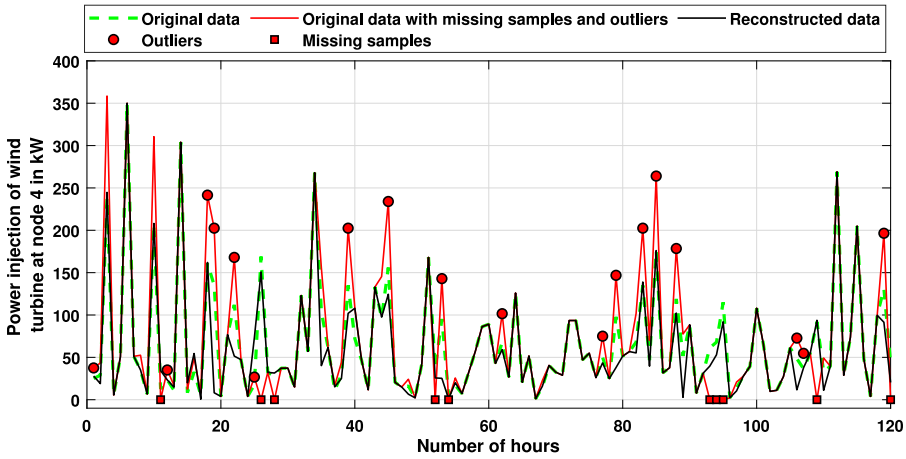


Fig. 10. Wind turbine power injection at node 4 (5 days data) of test system-2 with missing samples and outliers.

existing measurement data are applied to the DSSE algorithm to get an accurate solution.

4.3. Identification of topology with the presence of missing samples and outliers

In this section, the dense layers are connected by excluding the decoder part of the trained DAE, CDAE, LDAE, and TCDAE models for performing the multi-label classification task. Eventually, the models are again trained in a supervised manner by applying the measurement data along with its binary output of a particular topology. The training of the models is carried out by k-fold validation with 5-folds. The effect of incomplete data and outliers on TI is evaluated using the classification metrics and the testing performance of the DAE models with these metrics is mentioned in Table 3. The formulas of the evaluation metrics in the classification task are given by

$$Accuracy = \frac{TP + TN}{TP + FP + TN + FN} \tag{10}$$

$$Precision = \frac{TP}{TP + FP} \tag{11}$$

$$Recall = \frac{TP}{TP + FN} \tag{12}$$

$$F_1 - Score = 2 \times \frac{Precision \times Recall}{Precision + Recall} \tag{13}$$

where, TP = True Positive, TN = True Negative, FP = False Positive and FN = False Negative.

The classification metrics such as accuracy, precision, recall, and the F_1-score (which represents the geometric mean of the precision and recall metrics) are considered for evaluating the accuracy and preciseness of the models in TI. From Table 3, the high values of accuracy and F_1-score clearly indicating that the proposed TCDAE model is efficient in classifying the topologies accurately and precisely. Since the LSTM-based DAE model is effective in capturing the temporal features from the real-time hourly data, it is occupying the second-best model. The accuracy of the DAE models considered in this work is not much affected by the increase in the percentage of missing measurements. However, the loss of half of the measurements at once does not occur in practice, and thereby the proposed TCDAE model can be used in real-time after training the model with all possible scenarios for each topology. For instance, the number of correct identification of topologies with the testing dataset can also be described using the confusion matrix and it is shown in Table 4 for the proposed TCDAE model with 10% missing and 30% outliers. The structure of confusion matrix is represented in Fig. 11. From the Table 4, it can be clearly demonstrated that the proposed model is more accurate in identification of topologies as the

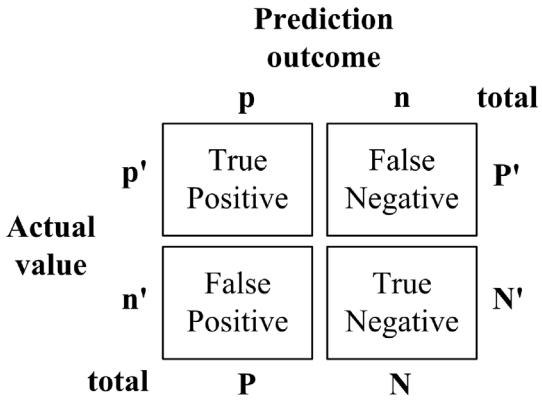


Fig. 11. Confusion matrix.

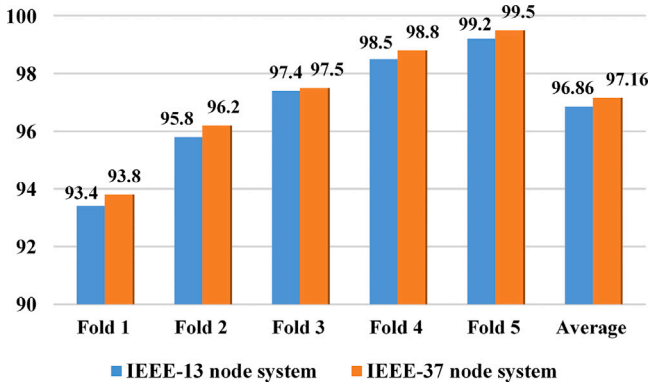


Fig. 12. 5-fold validation accuracy in % of TCDAE model with 10% missing and 30% outliers.

number of true positive and true negative predictions are very high for all labels. Also, the 5-fold validation is performed on the training dataset to eliminate a biasing problem and it facilitates proper training of the model for all datasets scenarios. Each fold validation accuracy of the proposed TCN based DAE model is shown in Fig. 12.

The computational times of different DAE-based models for TI with training and testing cases are represented in Table 5. In general, the time taken by the simple DAE model is minimum compared to convolutional DAE and LSTM-DAE. Since the TCDAE model can extract both spatial and temporal features with dilated causal convolutions, the model needs some extra training time compared to CDAE and LDAE

Table 3
Testing metrics in % of different DAE-based DL models.

Testing metrics with different % of missing and 30% outliers						
Test system	Missing %	Metric	DAE model	CDAE model	LDAE model	TCDAE model
Test system-1	10%	Accuracy	94.74	95.06	95.32	**97.30**
		Precision	95.57	95.88	96.08	**97.69**
		Recall	95.53	95.77	96.02	**97.75**
		F1-score	95.55	95.83	96.05	**97.72**
	30%	Accuracy	91.72	92.28	92.76	**95.40**
		Precision	93.25	93.74	93.96	**95.96**
		Recall	92.72	93.18	93.79	**96.28**
		F1-score	92.98	93.46	93.88	**96.12**
	50%	Accuracy	85.99	86.46	87.05	**89.98**
		Precision	88.38	88.63	89.13	**91.73**
		Recall	87.88	88.47	88.98	**91.31**
		F1-score	88.13	88.55	89.06	**91.52**
Test system-2	10%	Accuracy	96.11	96.95	97.15	**98.49**
		Precision	96.55	97.30	97.47	**98.54**
		Recall	96.26	97.06	97.27	**98.69**
		F1-score	96.40	97.18	97.37	**98.61**
	30%	Accuracy	94.87	95.21	95.32	**97.20**
		Precision	95.12	95.42	95.52	**97.32**
		Recall	95.43	95.75	95.86	**97.52**
		F1-score	95.27	95.59	95.69	**97.42**
	50%	Accuracy	87.85	88.32	88.65	**91.54**
		Precision	88.44	89.22	89.50	**92.00**
		Recall	89.25	89.23	89.56	**92.43**
		F1-score	88.84	89.22	89.53	**92.21**

Table 4
Confusion matrix of TCN-based DAE model with 10% missing and 30% outliers.

		Test system-1		Test system-2	
		p	n	p	n
1st label (MSB)	p′	1701	0	3615	10
	n′	3	216	13	2362
2nd label	p′	934	2	3555	19
	n′	6	978	27	2399
3rd label	p′	907	45	3028	46
	n′	48	920	36	2890
4th label	p′	902	55	3093	31
	n′	48	915	21	2855
5th label (LSB)	p′	–	–	2750	107
	n′	–	–	141	3002

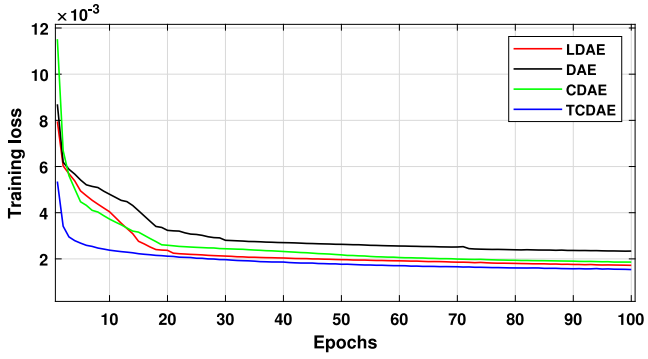


Fig. 13. Convergence plot of training loss.

models. The convergence plot of all DAE models is depicted in Fig. 13. Based on the convergence plot of training loss, the proposed model's convergence is achieved with minimum number of epochs compared to other DAE models. Thus, the TCDAE model can be trained with less number of epochs, which leads to minimum computational training time compared to other DAE models.

Although the training process requires some extra time, the computation times in the online testing stage are very low for all the DAE models (<1 s). Since the accuracy of the proposed TCDAE model for TI is quite high compared to other DAE models, the proposed model can be used in real-time even with the expense of a small fraction of time.

4.3.1. Case 1: Topology change after the removal of 2–7 branch in the test system-1

The case study is carried out by assuming that a fault occurred in a branch 2–7 and accordingly, the 1st topology has been changed to 8th topology by closing the tie-line switch of branch 11–7 to avoid islanding of node 7. With this condition, the topology processor has to detect the change in topology immediately otherwise the solution of the state estimation gets affected. The Table 6 shows that the corresponding simulation results with and without TI and they are evaluated using a two-norm error metric which is given by [30],

$$\text{Two-Norm Error} = \sqrt{\sum_{j=1}^{M}\sum_{k=1}^{3}\left|v_{j,k}^{est} - v_{j,k}^{act}\right|^2} \tag{14}$$

where M is number of nodes in the system, $v_{j,k}^{est}$ is the jth node, k^{th} phase voltage magnitude estimate in pu and $v_{j,k}^{act}$ is the jth node, k^{th} phase true voltage obtained from power flows in pu ($\varepsilon_{|V|}$). The two-Norm error is also calculated for voltage phase angles in radians (ε_{δ}).

4.3.2. Case 2: Topology change after the removal of branch 3–14 in the test system-2

In this case, the fault is created in a branch 3–14 and accordingly, the 1st topology is changed to 2nd topology after closing the tie-line switch of branch 32–26 for avoiding the islanding of the system. The corresponding simulation results with and without TI are represented in Table 6.

Based on the results, it can be concluded that the two-norm error has been increased if the topology processor does not respond immediately and the DSSE algorithm runs with the same topology that exists before the removal of the line. Hence, the error in node voltages increases which results in the overall two-norm error of the system is getting

Table 5
Computational times of different DAE-based DL models for TI.

Model	Test system-1		Test system-2	
	Training time in sec	Testing time in sec	Training time in sec	Testing time in sec
DAE	120	0.08	261	0.12
CDAE	168	0.1	418	0.18
LDAE	205	0.16	525	0.21
TCDAE	250	0.20	602	0.25

Table 6
Two-norm error of the test cases.

	Test system-1		Test system-2	
	$\epsilon_{\|V\|}$	ϵ_{δ}	$\epsilon_{\|V\|}$	ϵ_{δ}
Estimation with TI	2.82×10^{-2}	1.11×10^{-2}	6.2×10^{-2}	0.63×10^{-2}
Estimation without TI	9.9×10^{-2}	8.8×10^{-2}	17.6×10^{-2}	3.76×10^{-2}

increased as shown in Table 6. Therefore, the accurate topology identification and the measurement data without any missing samples and bad data are very much essential for the DSSE algorithm to provide an accurate solution.

5. Conclusion

In this paper, different denoising autoencoder-based deep learning models are implemented for accurate topology identification with the existence of missing data and outliers. The proposed temporal convolutional DAE model with modified Huber-loss function is a novel DL approach and its performance is evaluated with different percentages of missing samples and outliers. For this work, the modified IEEE 13-node and IEEE 37-node unbalanced distribution systems are employed by considering the real-time datasets of various types of loads such as residential, commercial, and industrial loads. Furthermore, the real-time datasets of renewable energy sources are also employed to study the performance of the proposed model with timely varying penetration levels of RES. Since the modern line current sensors are non-contact and inexpensive, the line current measurements along with the power injection measurements are employed for the analysis. The efficacy of the proposed model in comparison with other DAE-based models is evaluated for both the reconstruction of incomplete data and outliers, as well as the classification of topologies. The reconstruction results are evaluated with an RMSE metric and the classification results are demonstrated with accuracy and confusion matrix. From these results, it can be concluded that the proposed TCN-based DAE is more robust to the uncertainties in measurement data such as missing data and outliers. In addition to this, the importance of accurate topology detection in DSSE is clearly explained with a two-norm error metric by conducting suitable case studies on the test systems.

CRediT authorship contribution statement

Y. Raghuvamsi: Conceptualization, Methodology, Software, Visualization, Writing – original draft. **Kiran Teeparthi:** Writing – review & editing, Supervision, Validation, Investigation. **Vishalteja Kosana:** Conceptualization, Validation, Data curation.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

Data availability

The authors do not have permission to share data.

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